

Highlights

Conformal Surprisal Fusion for Detecting Prompt-Injection Compromise in LLM Agent Traces

Quang-Vinh Dang

- Fuses observation- and action-side trace signals to detect agent compromise
- Conformal surprisal fusion is scale-free and needs no tuned signal weights
- Raises AUROC 0.937 to 0.954 and cuts recall variance by two orders
- Clipping at the benign maximum provably caps AUROC at one half
- AgentDojo hides almost no compromise, so it cannot test the hard case